Rajarata University of Sri Lanka

Department of Computing

COM 3401 – Data Communication and Networking
Laboratory 02 – Ethernet communication, Switching and ARP
Time: 2 Hours

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to:

1. Learning on MAC address, switching and ethernet communication
2. Getting familiar with ARP

Pre-Practical Preparation

1. **Reading/Resources**: Refer to the topics covered in Lecture 2 and Lecture 3

Practical Objective:

Explain how Ethernet communication take place. Explain how MAC address play a pivotal role in switching. Practically explain the difference between the functioning of switch and hub. Students will learn how Layer 2 communication take place and practicality of using a switch and a router.

Practical Activity

Activity 1:

Visualizing the packet flow in a LAN

- Objectives: Introduction to simulation mode in Cisco Packet Tracer
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 3 x PC
    3. Connect three PCs to the switch using Copper straight-through cables
    4. Configure PC0 with following IP configuration
        - IP: 192.168.1.10

- Mask: 255.255.255.0

5. Configure PC1 with following IP configuration
   - IP: 192.168.1.20
   - Mask: 255.255.255.0
6. Configure PC1 with following IP configuration
   - IP: 192.168.1.30
   - Mask: 255.255.255.0
7. Switch to simulation mode
8. Click on 'show All/None' and clear default filters
9. Click on edit filters and select ARP and ICMP underIPv4 tab
10. Open the command prompt of PC0 and ping PC1
11. Click the play button in the simulation mode and observe the packet flow and the command prompt side by side
12. Repeat the procedure with other PCs

Activity 2:

Introducing MAC address

- Objective: Introducing the concept of MAC address and making students familiar with the commands used to find the MAC address of the devices
- Instructions:
  1. Retain the project created in Activity 1
  2. Access the command prompt of PC 1 and type the following command
     - *Ipconfig /all*
  3. Note the address displayed ad 'Physical Address'
  4. Access the command prompt of PC 1 and type the following command
     - *arp -a*
  5. Go to the CLI of Switch and type following commands
     - *enable*
     - *show mac address-table*
  6. Find the purpose of two tables

Activity 3:

Demonstrating ARP

- Objective: Introducing the concept of APR and how MAC address tables are updated
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 4 x PC
    3. Connect PCs to the switch using Copper straight-through cables
    4. Configure the LAN
    5. Switch to simulation mode
    6. Display the ARP table of PC0 and PC2 Side by side
    7. Ping PC1 from PC0
    8. Observe the packet behaviour and how ARP table entries are created
    9. Try the same procedure with other combination of devices

Activity 4:

Switch Vs Hub

- Objective: Introducing the concept of forwarding intelligence
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x hub (Hub-PT)
        - 4 x PC
    3. Connect PCs to the hub using Copper straight-through cables
    4. Configure the LAN
    5. Switch to simulation mode
    6. Ping PC0 from PC0
    7. Observe the packet behaviour